

5.3.6.1 Current consumption in Run mode

Table 24. Typical and maximum current consumption in Run mode

Evaluated by characterization - not tested in production, unless otherwise stated.

Clocked by HSI at 144 MHz or HSIDIV3 at 48MHz if not otherwise specified.

Symbol	Parameter	Conditions	f _{HCLK} (MHz)	Typ	Max					unit
					T _J = 30°C	T _J = 90°C	T _J = 110°C	T _J = 130°C	T _J = 140°C	
I _{DD(Run)} ⁽¹⁾	Supply current in Run mode all peripheral clocks disabled	Code in Flash, ICACHE 2-Ways	144	11.5	12.00	13.50	14.50	17.00	18.50	mA
			48	4.1	4.35	5.75	7.15	9.45	11.00	
		Code in Flash, ICACHE OFF	144	9.4	10.00	11.50	12.50	15.00	16.50	
			48	4.0	4.35	5.70	7.05	9.35	11.00	
		Code in SRAM2, ICACHE OFF, FLASH ON	144	9.3	9.80	11.00	12.50	15.00	16.50	
			48	3.4	3.60	5.05	6.40	8.70	10.50	
	Supply current in Run mode all peripheral clocks enabled	Code in Flash, ICACHE 2-Ways	144 ⁽²⁾	23.5	24.50	25.50	27.00	29.50	31.50	
		Code in Flash, ICACHE OFF		21.0	22.50	23.50	25.00	27.50	29.00	

1. Measures done with prefetch enabled.

2. Clocked by PSI at 144 MHz with HSE at 16 MHz in bypass mode.

Table 25. Typical current consumption in Run mode with CoreMark running from flash memory and SRAM

Evaluated by characterization - not tested in production, unless otherwise stated.

Symbol	Parameter	Conditions	SYSCLK source	f _{HCLK} (MHz)	Typ	Unit	Typ	Unit
		Peripheral						
I _{DD(Run)} ⁽¹⁾	Supply current in Run mode	Code in Flash, ICACHE 2-WAY, prefetch ON	PSI ⁽²⁾	144	12.00	mA	82.00	μA/ MHz
		Code in Flash, ICACHE 1-WAY, prefetch ON			9.85		68.50	
		Code in Flash, ICACHE OFF, prefetch ON			9.45		65.50	
		Code in Flash, ICACHE OFF, prefetch OFF			8.40		58.50	
		Code in SRAM2, ICACHE 2-WAY			11.50		78.00	
		Code in SRAM2, ICACHE 1-WAY			9.35		65.00	
		Code in SRAM2, ICACHE OFF			9.15		63.50	

1. Measures done with prefetch enabled.

2. Clocked by PSI 144 MHz with HSE at 16 MHz on bypass mode.

5.3.6.2 Current consumption in Sleep mode

Table 26. Typical and maximum current consumption in Sleep mode

Evaluated by characterization - Not tested in production.

Clocked by HSI at 144MHz or HSIDIV3 at 48MHz.

Symbol	Parameter	Conditions	f _{HCLK} (MHz)	Typ	Max					Unit
					T _J = 30 °C	T _J = 90 °C	T _J = 110 °C	T _J = 130 °C	T _J = 140 °C	
I _{DD(SLEEP)}	Supply current in Sleep mode	All peripheral clocks disabled	144	2.05	2.35	3.95	5.50	8.05	10.00	mA
			48	0.93	1.15	2.80	4.35	6.95	8.85	
		All peripheral clocks enabled	144	15.00	15.50	17.00	18.50	21.00	22.50	
			48	5.30	5.60	7.00	8.40	11.00	12.50	